Worksheet 4 · Factorising & Quadratics IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Factorise expressions by taking out common factors • Factorise quadratics, including the difference of two squares • Solve quadratic equations by factorising and by using the formula		
Grade: / 30	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Factorising & Quadratics*

Question 1 (2 marks)

Factorise fully $9ef - 12f$

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(2)

Question 2 (2 marks)

Factorise fully $5y + 20y^2$

(2)

Question 3 (2 marks)

Factorise fully $16c^4p^2 + 20cp^3$

.....

(2)

Question 4 (1 mark)

(c) Factorise $4c^2 - 9d^2$

.....

(1)

Question 5 (2 marks)

Factorise fully $2e^2 - 18$

.....

(2)

Question 6 (2 marks)

Factorise $2x^2 - 7x + 6$

.....

(2)

Question 7 (2 marks)

Factorise fully $25a^4c^7d + 45a^9c^3h$

.....

(2)

Question 8 (2 marks)

Factorise fully $15b^5c - 35b^3c^9$

.....

(2)

Question 9 (2 marks)

Factorise $y^2 - 2y - 48$

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(2)

Question 10 (2 marks)

Factorise $a^2 - 144$

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(2)

Question 11 **(2 marks)**

Factorise $9x^2 - 4y^2$

.....

(2)

Question 12 **(2 marks)**

Factorise $6y^2 - y - 5$

.....

(2)

Question 13 **(2 marks)**

$3x^2 - 8x - 3$

.....

(2)

Question 14 **(2 marks)**

Factorise completely $(12x - y)^2 - (4x - 3y)^2$

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Question 15 **(3 marks)**

(d) (i) Factorise $x^2 + 2x - 24$

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(2)

(ii) Hence, solve $x^2 + 2x - 24 = 0$

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(1)